

EDGE GESTURE PREDICTION Using Artificial Intelligence

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Artificial intelligence for gesture detection and classification has transformative potential across industries, from healthcare to manufacturing and consumer devices. It enables tasks like motion tracking in smartwatches, anomaly detection in machinery, and vehicle predictive analysis. Implementing AI on compact devices like the Indus-Board Coin—a 3cm innovation with built-in sensors—demonstrates how these capabilities can be brought to edge devices with limited resources.

We create a simple model for gesture detection, designed to classify motions commonly used in smartwatches, phones, and exercise-related devices. The model is optimised to run efficiently on a microcontroller (MCU), overcoming constraints like limited RAM and storage while delivering reliable AI performance.

Only a few components, detailed in the accompanying table, are needed to build the device. Fig. 1 shows the prototype testing setup, showcasing how AI can seamlessly execute tasks on the edge.

An online tool named Edge Impulse was used to collect the data, train a machine learning model, and deploy it on the IndusBoard MCU. An Edge Impulse account must be created using the link <https://edgeimpulse.com>. The developer and

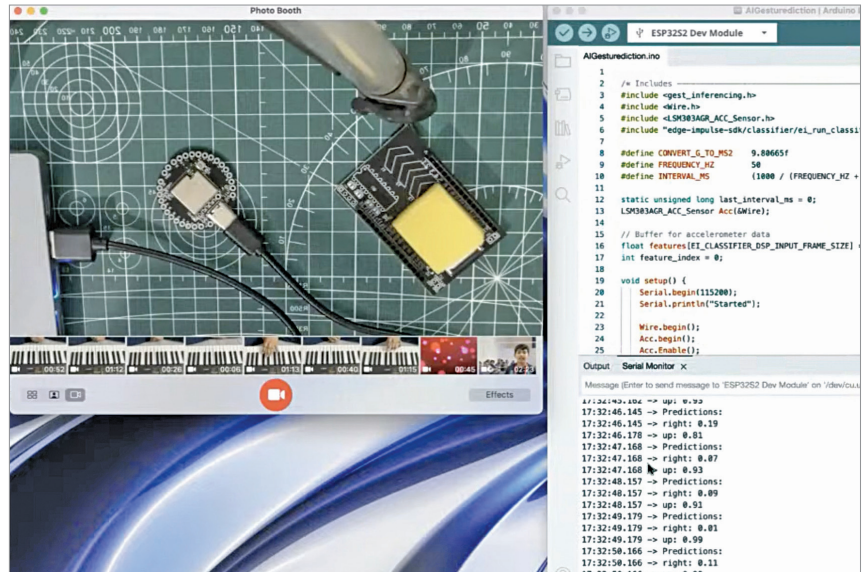


Fig. 1: Prototype testing setup

```

Allsensordataforwarder.ino
1  #include <Wire.h>
2  #include <LSM303AGR_ACC_Sensor.h>
3  ///////////////edge-impulse-data-forwarder --clean/////
4  #define CONVERT_G_TO_MS2  9.80665f
5  #define FREQUENCY_HZ      50
6  #define INTERVAL_MS        (1000 / (FREQUENCY_HZ + 1))
7  static unsigned long last_interval_ms = 0;
8  // Components for the LSM303
9  LSM303AGR_ACC_Sensor Acc(&Wire);
10 void setup() {
11     Serial.begin(115200);
12     Serial.println("Started");
13
14     // Initialize I2C bus.
15     Wire.begin();
16
17     // Initialize LSM303 accelerometer
18     Acc.begin();
19     Acc.Enable();
20     Acc.EnableTemperatureSensor();
21
22     Serial.println("LSM303 Accelerometer initialized");
23 }
24 void loop() {
25     int32_t accelerometer[3];
26     if (millis() > last_interval_ms + INTERVAL_MS) {
27         last_interval_ms = millis();
28         // Read accelerometer data
29         Acc.GetAxes(accelerometer);
30         // Convert to m/s^2 and print
31         Serial.print(accelerometer[0] * CONVERT_G_TO_MS2 / 1000.0f); // Converting from mg to m/s^2
32         Serial.print('\t');
33         Serial.print(accelerometer[1] * CONVERT_G_TO_MS2 / 1000.0f); // Converting from mg to m/s^2
34         Serial.print('\t');
35         Serial.println(accelerometer[2] * CONVERT_G_TO_MS2 / 1000.0f); // Converting from mg to m/s^2
36     }
37 }
38

```

Fig. 2: Edge Impulse data forwarder code

BILL OF MATERIALS

Components	Quantity
IndusBoard Coin	1
USB C-type cable	1